



PRONOE

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

PRONOE

Company or organization location (we welcome applicants from anywhere in the world)

Paris, France

Name(s) of primary point(s) of contact for this application

Nicolas Sdez and Juan Buceta

Brief company or organization description <20 words

PRONOE restores the ocean's natural capacity to permanently remove CO₂ from the air, in a sustainable, scalable and infrastructure-light manner.

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

Oceanic CO₂ capture and storage

Atmospheric CO₂ levels are naturally regulated by the dissolution of CO₂ in the ocean. As CO₂ dissolves, the surface layer of the ocean becomes more acidic [1]; this acidity is slowly transported to the deeper ocean layers over approximately 1,000 years, where it reacts with CaCO₃ deposits on the ocean floor [2]. This process dissolves CaCO₃ and restores the ocean's pH balance. Currently, CO₂ emissions from human activities are not counterbalanced by natural ocean regulation, resulting in increased atmospheric CO₂ concentration and ocean acidification.

The CO₂ equilibrium between the ocean and the atmosphere takes less than one year across most of the planet [4]. Consequently, CO₂ could be captured from the atmosphere within this time frame by alkalizing the ocean surfaces, a process known as Ocean Alkalinity Enhancement (OAE). Once captured, CO₂ remains stored in the ocean as bicarbonates (HCO₃⁻) for 10,000 to 1,000,000 years [5].

Chemistry behind oceanic CO₂ capture and storage

Seawater alkalinity or Total Alkalinity (TA), is defined as the capacity to resist changes in pH that would make the water more acidic [6]. Increasing TA does not necessarily mean removing CO₂. The dissolution of CO₂ in seawater involves 3 carbonated chemical species (i.e. CO₂, HCO₃⁻, and CO₃²⁻) [7]. The addition of alkalinity as carbonates (CO₃²⁻) to seawater would modify these equilibria and consume dissolved CO₂ according to the following reaction [8], [9]:



The addition of alkalinity as hydroxyls (OH⁻) to leads to:



Ultimately, the CO₃²⁻ obtained in Equation 2 entails the same CO₂ removal as Equation 1.

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

Best-in-class, differentiated technology and approach

OAE is broadly accepted as being a high-quality and **high-potential** (100 GtCO₂/y) approach that entails **fast capture** (<1 y) and **long-term storage** (>10,000y).

The principle of OAE consists of **enhancing oceanic, abiotic, natural carbon cycles** – whereby atmospheric CO₂ is captured and sequestered in seawater as dissolved inorganic carbon. This has **direct business implications**: no reliance on a third party for geological storage (**vertical integration**) and **broad geographical compatibility** – typically, any coastal or offshore location can be addressed, providing more operational freedom than most capture-only technologies (e.g., DAC/DOC).

Despite unique **environmental co-benefits** through ocean and/or coastal acidification mitigation, the effective development of OAE approaches has been **historically hindered by three *different but interconnected* issues**: 1) **availability of alkaline minerals**, 2) **disposal of acid wastes**, and 3) **permitting**.

1) **Highly alkaline minerals**, either naturally occurring (olivine, basalt) or man-made (slacked lime), typically considered **for Mineral OAE** (i) are unevenly distributed globally, (ii) are not extracted nor produced at a scale consistent with meaningful climate action, (iii) may contain environmentally-harmful heavy metals such as Iron or Nickel (Fe, Ni), (iv) may require energy-intensive thermal and/or mechanical processing. .

2) Instead of adding alkalinity, **electrochemical-based OAE** aims to remove acidity from the ocean, thus generating **large volumes of *dilute* acidic brines**, challenging to valorize, or dispose of.

3) Most approaches are based on “rapidly dissolving” or Dissolved **hydroxyl-based alkalinity** (OH⁻), entailing uncertainties and/or permitting difficulties related to turbidity levels, large pH variations, dissolution rates, secondary precipitation.

PRONOE tackles all three issues through its holistic approach for Sustainable Scalability and unique technology.

OAE takes many forms, PRONOE, uniquely pursues **Dissolved Carbonate OAE from Coastal Outfalls**.

OAE from **Coastal Outfalls** is deemed the single best OAE approach for focused MRV, providing **high Verifiability** as exemplified by Isometric’s choice of this pathway to set the first Ocean CDR protocol. Leveraging legacy coastal outfalls, massive available water flows, and **operating within the scope of in-force permits** ensures we work within the highest **Safety and Legality** standards, while actively **maximizing environmental and industrial co-benefits**, which ultimately contributes to **rapid deployability**.

In a **paradigm shift**, PRONOE has developed a patented process for dissolved carbonate alkalinity production with fully integrated, ex-situ acid neutralization. Our **zero-waste**-generating systems use **cheap, ubiquitous, unground, mildly alkaline mineral feedstocks**, alleviating mineral feedstock scalability issues.

In practice, we develop and deploy automated water treatment systems that are seamlessly integrated downstream of the operations of coastal industrial sites with a return flow to the sea. These systems

safely make returns flows less acidic, or more alkaline, within applicable regulations. See PRONOE's "BlueBox" in Figure 1.

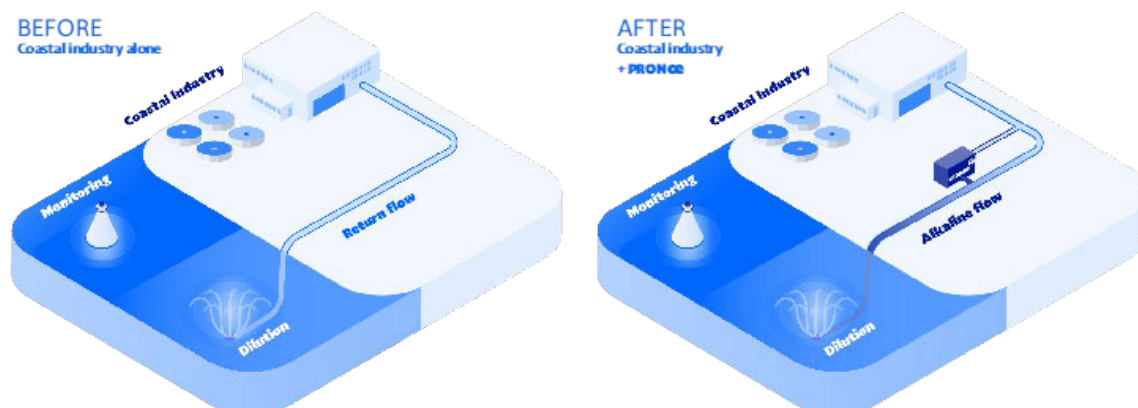


Figure 1 – PRONOE's co-location approach.

Based on electro-membrane technology, our systems are extremely compact, have a **low physical footprint** (e.g. less than 30mx30m for a 10 ktCDR/year capacity system), and are installed on previously artificialized land. These systems require only electricity and operate at ambient pressure and temperature with **low power consumption at scale**.

Modular technology allows for ease of scale-up. **Automated systems** and no need for downstream CO₂ logistics allow for **fully decentralized** CO₂ removal activities. Distributed PRONOE **systems of 10 ktCDR/year and 50 ktCDR/year** lead to **easy-to-finance, rightly-sized projects** and ultimately to competitive economics thanks to scale *and* learning effects.

Hardware standardization coupled with software-enabled operational flexibility is at the core of our approach. Our **tolerant** process accommodates **varying mineral and water feedstocks** to best leverage local resources. The operational flexibility of our standardized systems enables us to co-locate with almost any industrial site with a return flow to the sea. This return flow can be an effluent, in the case of wastewater treatment plants or desalination plants, or cooling flows, in the case of seawater-cooled power plants, data centers, etc. Finally, the most electro-intensive step of our process runs in batches within a continuous process, enabling us to best use low-carbon or **intermittent renewable electricity**, off-peak hours.

Differentiation

Pathway differentiation.

To the best of our knowledge, PRONOE, a **XPrize Top 100** CDR startup, is the first globally to pursue **Dissolved Carbonate-based (CO_3^{2-}) Ocean Alkalinity Enhancement (DC-OAE)** from Coastal Outfalls.

Technological differentiation results from process-level design and optimization and mastery of neutralization kinetics – it includes:

- **Fully integrated, ex-situ acid neutralization step, uniquely accommodating cheap, ubiquitous mildly alkaline minerals** that are not *readily* applicable to any other CDR approach. See Application Supplement: Alkaline Feedstock, 3. & 4.
- **Unique electro-membrane design, material and operation.** With over 30% reduction of membrane area and water consumption, as well as over 75% reduced pre-treatment intensity.

These differentiators impact key performance metrics, including:

- Significantly higher alkalinity addition (2x-7x) at iso-pH variation (carbonates vs. hydroxyls), allowing meaningful CDR activities with applicable regulations
- Significantly reduced CapEx and Electricity consumption for electro-membrane technologies
- Zero-acid-waste generation, and no heavy metal release
- Drastically reduced process water volumes, allowing for compact equipment, pumps, tanks, and overall a footprint allowing industrial integration
- High to Very High CDR net-negativity ratios (75%-95%)

Priority Innovation Areas

We trust this project addresses most if not all **Priority Innovation Areas for the 2025 prepurchase and Marine Carbon Removal**:

- Alignment with Frontier stated *mCDR's role in a CDR portfolio*
 - Shore-based project, leveraging on an abiotic mCDR pathway
 - Responsible deployment, including permitted operations within in-force permits, local government and scientists support, third-party monitored water parameters.
 - Unique approach for DC-OAE taking only the best, respectively of traditional mineral and electrochemical approaches: ease of monitoring and reduced environmental concerns, with credible feedstock sourcing strategy.
 - We believe that the use of limestone is mandatory, as a functionally unlimited feedstock to scale OAE globally, and that such use doesn't forcibly require low-cost, low-carbon MgO or CaO. At PRONOE, we directly functionalize limestone for mCDR,

right at the dispersion site.

- *Compelling evidence of co-benefits*

In parallel with this project, PRONOE has set up a collaboration with a leading marine institute to assess the impact of carbonate alkalinity enhancement on the early life stages of oysters. PRONOE will actively collaborate with our local industrial partners (20+ years of experience in brine valorization) to qualify further *identified industrial co-benefits*.

- *Robust, rigorous MRV*

The financing of robust, rigorous MRV is one of the key reasons we apply to Frontier RFP. We selected top-tier MRV partners and will follow best-in-class practices, including the requirements from Isometric protocols.

- *Integration with existing infrastructure*

A first of its kind in Europe, this proposed project will be taking place at an industrial desalination plant (4 900 m³/d) in the Canary Islands. Uniquely, this site will provide PRONOE with various brines and seawater feeds, contributing to proving process efficiency in the context of other industrial sites (wastewater, power plants, etc).

- *Robust feedstock sourcing and deployment strategy*

Tolerance of our operations to varying mineral feedstocks and water characteristics is a key outcome of this project. We will run our process with (bio)minerals of various physico-chemical characteristics to confirm local sourcing possibilities for future projects. If confirmed, potential to use (non thermally nor mechanically processed) limestone for mCDR (not ocean storage) would be a first.

- *Robust management of byproducts and contaminants*

Zero-waste-generation has been a self-imposed requirement since day one; we pursue the most feedstock-scalable, fully integrated, ex-situ acid neutralization strategy.

- *Resilience against immature supply chains*

Our process innovation relies on existing, industrial grade and scale components and equipment. Like Frontier does, we believe this criteria is of the utmost importance. See Annex 009 for details on PRONOE's supply chain.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale

criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

What PRONOE is building

PRONOE develops **compact, automated, feedstock tolerant, land-based industrial water treatment systems** that turn the return flows of coastal industries equipped with ocean outfalls into a **safe alkaline flow**, in **full compliance with applicable regulations**. This safe alkaline flow locally mitigates the acidification of surface waters, offering potential environmental co-benefits. As described in Equation 2 and Equation 3 (section 1.a), the increased alkalinity restores the seawater chemistry by converting dissolved CO₂ into bicarbonates (HCO₃⁻). The lower CO₂ levels in seawater trigger CO₂ capture from the air. Carbon is stored as bicarbonates for 100,000 to 1,000,000 years [5].

More fundamentally, we are building an infrastructure-light, sustainable, scalable and fully decentralized approach to **single-step CO₂ removal** (capture+storage; CDR), that can adapt to any coastal geography **worldwide** and bring **low-cost, high-quality CDR**, while **benefiting local industries, communities and ecosystems**.

This project: Location & scale

For this project, we will consist in the deployment of **a commercial demonstrator removing more than 100t/year of CO₂ (net)**. This RFP will finance the field-testing and demonstration of PRONOE's process, including fixed and variable OpEx, as well as best-in-class Carbon and Environmental Monitoring, Reporting and Verification (cMRV/eMRV) expenses. Site selection is paramount for a successful first commercial deployment; we selected an exceptional location for a **first-of-a-kind Ocean Alkalinity Enhancement (OAE) from Coastal Outfall in the EU**. Our project takes place in a world-leading hub for ocean sciences and desalination industry innovation, with strong political will and community acceptance. The **pre-permitted alkalinity addition** will occur at a contracted **industrial (4 900 m³/d) desalination plant in the Canary Islands** (Spain, Europe), offshore West Africa. This location is disproportionally affected by water scarcity; with more than 50% of their water supply coming from seawater desalination, the Canary Islands have the highest density globally of desalination plants. Moreover, ocean acidification is particularly acute in this area.

Our alkalinity addition is done according to applicable legislation, achieving conservative levels of alkalinity in the dispersed flow [10]. Our systems will be located in a barren area within the industrial complex (Figure 2 and Figure 22). That desalination site is a **(net) renewable energy producer** (wind and solar farms; Figure 2 and Figure 22). We can pioneer fighting ocean acidification in Africa, while leveraging EU grant support and an advanced, established ocean observation network (Copernicus/Mercator). This project will help us build

² We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

relevant knowledge to deploy in APAC, the Middle East, Africa and South America.



Figure 2 - Aerial view of the site, and picture of one PRONOE container.

Further pictures of the site are presented in Application Supplement: Ocean and Inland Water. The process diagram is presented in section 2a (Figure 4).

Beyond this project: Location & scale

At scale, PRONOE's industrial systems have a gross CDR capacity of **10kt or 50 ktCO₂/year**, seamlessly integrated downstream of coastal industrial sites and utilities, installed on their artificialized grounds. These coastal sites include **desalination plants, wastewater treatment plants, and power plants**. Our view of pragmatic OAE from coastal outfalls relies on multiple, mid-sized industrial systems in various locations, as opposed to monumental green-field and dedicated CDR plants. This mimics the deployment strategy of desalination plants; desalination has been one of the industries with the fastest-declining costs in the last 30 years. Beyond physical integration of our systems on site, PRONOE integrates with **existing value chains, supply chains and distribution networks**, enabling new circular economy business models.

Today, the **desalination industry** disperses more than 25 Gm³/year of brines globally, enabling more than **200 MtCDR potential**, as confirmed by RMI Report [11]. Desalination is projected to grow at a rate of 7.5 to 9.5% per year. The return flows to the sea of wastewater treatment plants (WWTP) and (nuclear) power plants are 8x and (2x) 40x larger, respectively.

Driven by urbanization and electrification, return flows to the sea from coastal wastewater treatment plants and nuclear power plants are expected to be multiplied by 3x and 2x, respectively, by 2050. **By the mid-century, each of these utility verticals would have the potential to enable Gt-scale CDR.**

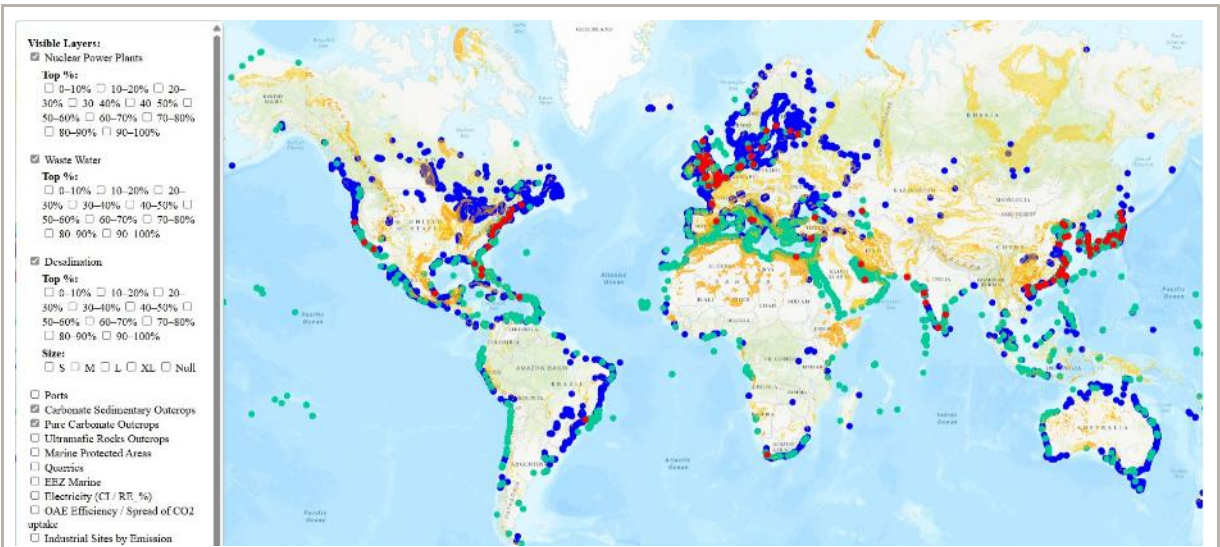


Figure 3 - In-house map showing locations of Wastewater Treatment (blue), Desalination (green), and Nuclear Power (red) plants (overlaid in that order) in the vicinity of coasts and major waterways.

Current cost breakdown

This project aims at **upgrading a technical (process) demonstration into a commercial (dispersion) demonstration**, hence the cost structure is highly specific, with 50%+ of costs related to MRV. See TEA spreadsheet, and section 6 for more details.

Approaching <100\$/t cost and >0,5Gt scale criteria

We align with Frontier’s removal capacity target (0.5 Gt), aiming at removing 0.7+ Gt/year. This will require a supply chain effort for scaling up the manufacturing and assembly capacity of currently available industrial equipment, yet, importantly, face *no physical limitations for hardware manufacturing or feedstock consumption*.

The cost breakdown at different scales is described in the techno-economic analysis (Excel) file. The costs, and the price per ton proposed for this project of 1499 \$/tCDR are similar to other early-stage Ocean CDR projects and explained by the absence of scale effects and process and energy inefficiencies at a small scale.

Based on suppliers’ quotations, third-party validated laboratory tests, and today’s feedstock prices, combined with industry-standard scaling and learning effects, we are confident in reaching a cost below 150 \$/tCDR at Mt scale.

Meeting Frontier’s removal price target (<\$100) necessitates a further cost decline of electricity from the conservatively assumed \$60/MWh to \$40/MWh (-33%), as well as a 20% decrease in the cost of mineral feedstock and transportation.

CDR Quantification

Our MRV approach consists of several key components with varying measurement and modeling requirements depending on temporal and spatial scales (Figure 14).

Our approach is **consistent with best-in-class methodologies as set out by recently released independent standards**. After a positive preliminary, third-party project risk assessment, see Annex 004, we implemented additional risk minimisation measures, including securing carbon rights, and benchmarking and selecting appropriate region ocean models.

First, return flows are monitored in land, upstream and downstream of our system to precisely monitor dissolved alkalinity addition. Second, the mixing and extent of the plume mixing zone downstream of the outflow are both modeled and measured. This serves to determine the alkalinity forcing function with a regional model or with sensitivity analysis in the air-sea gas exchange model. Third, the alkalinity forcing is used in a regional or global air-sea gas exchange model to quantify gross CDR, with historical data serving as a baseline. Lastly, various biogeochemical feedbacks that could lead to alkalinity losses are quantified, including biotic calcification response and secondary precipitation.

Our vision for developing robust MRV relies on two pillars:

(i) Collaborations with MRV actors in the private sector; being their first OAE project outside North America and connecting them to the European CDR ecosystem.

(ii) Research with local public centers. The selected site offers exceptional MRV characteristics given the previous environmental research in that very same coastline. PRONOE has been in contact with local actors working on the region, who have already gathered historical data to serve as a baseline [15], and mathematical models of the near-field that have been validated. The environmental response was also studied in the region, which helped fast-tracking permitting and will help minimize ecological risks. Experimental results proved the decrease in pCO₂ due to our alkalinity addition, which will be measured with fixed sensors at the dispersion points.

PRONOE's MRV approach is detailed in section 5.

Quantifying and qualifying environmental and social impacts

Ocean acidification impacts on natural ecosystems and human activities have been extensively discussed, namely the fact that critically endangered coral reefs provide food for half a billion people globally. Third-party, ex-situ ecotoxicity and ecological tests, as well as scientifically overviewed, responsible field deployments are key. As we are not aiming only for social license, but for active community support. To that end, we published several expert committee-reviewed articles on the environmental impact of desalination-coupled OAE, and we will deploy best-in-class Whole Effluent Toxicity measurements based on the paradigm-shift direct, live monitoring of sentinel species' physiological status.

We are founding members of a global NGO (Water Positive) empowering local communities to become active stewards of sustainable water management and restoration. Working on this initiative, we understood the impacts and challenges of the communities disproportionately impacted by water scarcity. We identified synergies with local stakeholders to bring health benefits to coastal communities suffering desertification and water scarcity in many geographies in Africa, Europe, the Middle East, and South America. Additionally, our first deployment takes place in a highly touristic area, which will help us raise awareness about the role and importance of OAE as part of climate action.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

PRONOE has a solid risk management culture, grounded in Aerospace Industry-grade Risk Breakdown System. Before risk mitigation, rigorous risk minimization is performed. Specifically for this project a third-party risk assessment was performed (Annex 004), and key learnings implemented (Annexes 005–006–007).

Technical risks

Technical risks have been largely mitigated by two generations of lab-scale pilots, and a soon-to-be-deployed technical demonstrator. There are risks associated with scaling any technology. We mitigate these risks by starting from proven, industrial, and modular technological components. PRONOE scales the process, not the individual technologies or unit operations, and the main technical risk lies with the integration of the unit operations.

For this project, technical risk lies with the performance of the neutralization and capture reactors, namely with varying feedstocks qualities and characteristics. For example, low-mixing points in larger reactors lead to heterogeneities and lower conversions of the product of interest. Although validated thoroughly with long-term tests under harsh conditions and real-world brines with our industrial pilot in the lab, the steady performance of the membranes will be closely monitored, with automatic, instantaneous switch-off. Similarly, while never used in months in the lab, a Clean-In-Place equipment will be on site.

Project execution – Supply chain risks

PRONOE process combines existing technological components from various providers, with PRONOE acting as an OEM. This approach drastically mitigates technical risks, with standard systems engineering risks and interface management risks remaining, yet increases supply chain risks. Scaling our technology involves process design and optimization, unit operation specification, and project management to assemble, integrate, and test the system before delivery and on-site commissioning.

Based on our previous experience, we actively engage with industrial equipment and feedstock

providers to iterate quickly on specifications and functional requirements and select them after a structured process, including site visits. Supplier failure is further mitigated via clearly defined ARCI systems, systematic review of engineering deliverables, and regular meetings.

For this project, a binding quotation post-FEL-3 engineering has been delivered to PRONOE from an established supplier.

Project execution – Operating partner engagement risk

Our first approach for deployment involves co-locating our systems with existing coastal industries. Understanding the needs of the coastal industries is key to co-locating our activities and engaging them effectively and efficiently.

We successfully engaged with over 15 water treatment and energy corporations and documented their interest in our technology.

Successful interactions resulted in PRONOE issuing several studies and contracting a deployment site – putting PRONOE on track to being the first Ocean CDR project integrated into an industrial desalination plant in the EU. Our strategy includes a strong industry presence (e.g., IDA, WEX Global conferences and membership), and having secured top-tier global advisors in the industry. While our early focus is related to the desalination industry, we are actively exploring other verticals, including wastewater, power plants, and municipal and industrial cooling flows.

For this project, the site is contractually secured.

Monitoring, Reporting and Verification (MRV) risks

On one hand, our process locally increases the alkalinity of seawater enough to increase its CO₂ capture capacity. On the other hand, this alkalinity variation is not enough to be measurable with most remote-operated equipment (e.g., pH meters) and is hard to measure remotely in the environment (e.g., total alkalinity). Thus, the general consensus in the scientific community is that MRV shall consist of direct measuring and mathematical modeling.

At PRONOE, we believe it is important to build robust MRV methods within the community of ocean scientists. Risks are mitigated by having selected credible third-party partners to build our third-party-reviewed MRV approach with, including sensors and modeling, with redundancy from academic partners.

PRONOE's MRV approach is detailed in section 5.

Environmental risks

Preventing ecosystem risks is paramount and at the core of PRONOE's vision. Environmental risks left unresolved will likely (and rightfully) trigger social license risks. Firstly, we will strictly comply with the existing environmental regulations to ensure the safety of pH levels at the dispersion points. Local governmental actors and decision-makers on environmental issues reviewed the documentation

compiled in collaboration with a government-endorsed third party and approved the project.

Furthermore, the MRV and automation ensures safe operations through adapting our production based on pre-defined physico-chemical thresholds (e.g., pH) and continuous own and third-party measurement of both the discharge flow and the receiving environment.

We will leverage established ocean outfalls that have thoroughly validated their plume's mixing rates, thereby minimizing the affected area. Ecotoxicity tests have been carried out by third parties with our alkalinity flow (Annex 003). Additionally, in the context of this project, PRONOE will execute a best-in-class environmental monitoring plan, including the use of keystone species as biosensors through valvometry, IoT and AI.

We have several research programs ongoing, including biocalcification responses with IFREMER and the University of Alaska (NOPP). We are committed to closing knowledge gaps, and constantly updating with the information published by referents like OceanNets, Ocean Visions, HourGlass Climate on the environmental impact of OAE. In the context of this first project, we reached out to local oceanographers that have been studying for decades the carbon chemistry and biology of the region where we will deploy.

PRONOE's environmental risks are detailed in section Application Supplement: Ocean and Inland Water, points 4 and 5.

Market risks

The dynamics of the voluntary carbon credits market are uncertain, in total volumes, prices and quality expectations from purchasers. This is a risk, accounting for the relatively high price of our CDR certificates at the initial stages.

Navigating these markets over the next years and generating revenues through advanced market commitments will be key to scaling up our solution. Getting purchases from highly regarded and well-recognized buyers clubs such as Frontier will facilitate other early purchases, and may pave the way for catalytic offtake contracts. Through active participation in various leading CDR trade organizations, PRONOE is contributing to policymakers' awareness at European and international levels for the integration of Ocean CDR solutions in regulated markets. Being an industrially integrated CO2 removal solution is expected to help securing strategic partnership, and channel some or all of a project's removals for 'in-value-chain mitigation' purposes, making it comparatively less risky than stand-alone CDR projects.

- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	330 ton CO ₂
Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	Q3 2026 - Q4 2029
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	1426 \$/ton CO ₂
Levelized price (\$/ton CO ₂) ³ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	1499 \$/ton CO ₂ (2% discount from 5c, 3% margin)

³ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).